When VLMs 'Fix' Students: Identifying and Penalizing Over-Correction in the Evaluation of Multi-line Handwritten Math OCR

Jin Seong[1]*, **Wencke Liermann**[1]*, **Minho Kim**[1]†, **Jong-hun Shin**[1], **Soojong Lim**[1]
[1]Electronics and Telecommunications Research Institute, Republic of Korea
Correspondence: real_castle@etri.re.kr

Abstract

Accurate transcription of handwritten mathematics is crucial for educational AI systems, yet current benchmarks fail to evaluate this capability properly. Most prior studies focus on single-line expressions and rely on lexical metrics such as BLEU, which fail to account for the severity of differences. In this paper, we present the first systematic study of multi-line handwritten math Optical Character Recognition (OCR), revealing a critical failure mode of Vision-Language Models (VLMs): over-correction. Instead of faithfully transcribing a student's work, these models often "fix" errors, thereby hiding the very mistakes that educational assessment aims to detect. To quantify this phenomenon, we propose PINK (Penalized INK-based score), a semantic evaluation metric that leverages a Large Language Model (LLM) for rubric-based grading and explicitly penalizes differences that increase the grade. Our comprehensive evaluation of 15 state-of-the-art VLMs on the FERMAT dataset reveals substantial ranking reversals compared to BLEU: models like GPT-4o are heavily penalized for aggressive over-correction, whereas Gemini 2.5 Flash emerges as the most faithful transcriber. Furthermore, a human expert study shows that PINK aligns significantly better with human judgment (55.0% preference over BLEU's 39.5%), providing a more reliable evaluation framework for handwritten math OCR in educational settings.

Code: github.com/9115jin/PINK

1 Introduction

The deployment of Vision-Language Models (VLMs) in educational AI systems promises personalized learning at scale; however, it requires one fundamental capability: the faithful interpretation of students' handwritten work. Here, a key challenge is that, while existing Optical Character Recognition (OCR) research has matured around single-line expressions (Mouchère et al., 2016; Deng et al., 2017; Yuan et al., 2022), real student solutions span multiple lines, interweaving mathematical notation with explanatory text throughout the problem-solving process. As modern VLMs, equipped with remarkable capabilities in mathematical reasoning, autoregressively process this longer context, they build a strong, logically consistent internal narrative, which usually helps to solve ambiguities in the transcription process. However, when the model encounters an erroneous visual token (e.g., "5" in "2 × 3 = 5"), this token could create a conflict with the powerful internal context that

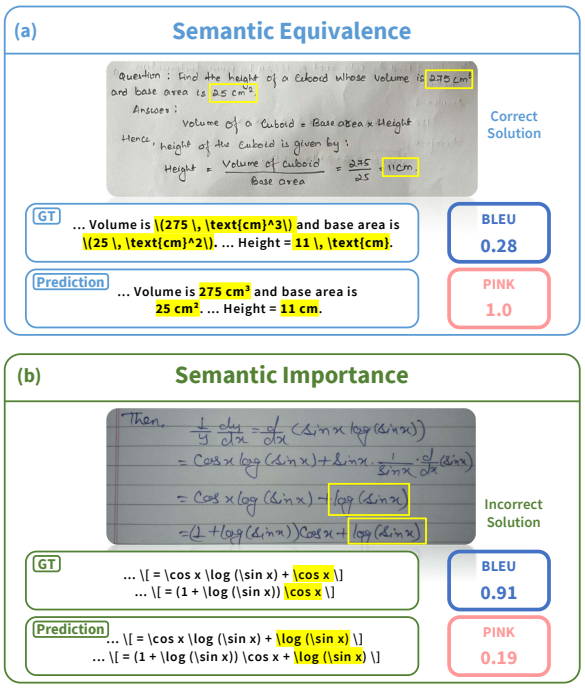

Figure 1: **BLEU's Blind Spots in Multi-line Handwritten Math OCR.** (a) BLEU penalizes semantically equivalent expressions. (b) BLEU assigns a high score even when the OCR misses a key part of the solution, treating a critical omission as insignificant.

* Equal contribution.
† Work done while interning at ETRI.

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implies the token in that spot must be ”6”. Consequently, a model might dismiss the visual evidence and generate a token that aligns with its logical reasoning (Li et al., 2025), erasing evidence of the student's misconception in the process, a phenomenon we call Over-Correction. Over-Correction has been observed across all modalities, including the fields of automatic speech (Udagawa et al., 2024) and grammar error recognition (Park et al., 2025).

Using the FERMAT (Nath et al., 2025) dataset, we quantify the phenomenon for handwritten multi-line math OCR, delivering evidence that Over-Correction is not limited to isolated sample instances but a pervasive failure mode of modern VLMs. The implications for educational AI are severe: an automated tutor receiving such "corrected" transcriptions would miss critical learning opportunities, potentially reinforcing misconceptions by failing to address them.

Our analysis employs PINK (Penalized INK-based score), a novel metric specifically designed to evaluate educational faithfulness. It leverages an LLM-based auto-grader with a carefully designed rubric system that assesses semantic mathematical correctness across five dimensions and allows for nuanced partial credit, i.e., 0-20 points, mirroring human grading. In a two-stage evaluation pipeline, PINK explicitly detects and penalizes over-corrections. It first computes an oracle baseline score by grading the ground truth transcription, before comparing it to the score assigned to the transcription generated by the OCR model. When a model's output scores higher than the oracle (indicating it "improved" the student's work), PINK applies graduated penalties, ensuring models are held accountable for transcription faithfulness. PINK also addresses another common deficit of traditional OCR evaluation metrics, e.g. BLEU (Papineni et al., 2002). It can recognize semantic equivalence and reflect semantic importance: a critical unit mistake (cm→m) receives more weight than minor formatting variations. As Fig. 1 illustrates, mathematically identical expressions like '\frac{1}{2}' and '0.5' are penalized by BLEU due to LaTeX formatting differences, while PINK correctly identifies them as expressions of equal meaning. To quantify this advantage, we analyzed 2,244 FERMAT (Nath et al., 2025) samples transcribed by Ovis2 8B VLM and found that 60.9% of discrepancies flagged by BLEU were limited to (8.2%) or included (52.7%) formatting differences.

We evaluated 15 SoTA VLMs (ranging from 7B to 124B parameters) using our framework, and the results revealed a complete departure from conventional BLEU-based rankings. Under BLEU, GPT-4o achieves 3rd place with 0.735; under PINK, it drops to 6th with 0.907, heavily penalized for aggressive over-correction. Conversely, Gemini 2.5 Flash rises from 10th (BLEU: 0.614) to 1st (PINK: 0.935) by maintaining exceptional transcription faithfulness. A human expert evaluation demonstrates that the ranking under PINK offers a more human-aligned perspective: given a choice between BLEU and PINK scores for the same OCR output, experts preferred PINK 55.0% of the time versus BLEU's 39.5%. Beyond rankings, our results also reveal a strong positive correlation between model scale and over-correction frequency, that is, larger models within the same family consistently exhibit higher over-correction rates, suggesting this behavior may be an emergent property of advanced reasoning capabilities (Wei et al., 2022).

In summary, our contributions are as follows.

- **First multi-line math OCR evaluation framework with semantic evaluation**: We adapt the FERMAT dataset to the task of OCR performance assessment and compare the abilities of 15 state-of-the-art VLMs.
- **Discovery and quantification of Over-Correction**: We identify and systematically analyze a critical VLM failure mode affecting 42.1%-66.2% of transcriptions, where models spontaneously "fix" student errors rather than faithfully transcribing them.
- **PINK metric validated by human preference**: We propose a faithfulness-aware metric that combines semantic accuracy with transcription fidelity, achieving 55% human preference over BLEU's 39.5% and revealing dramatic ranking reversals among state-of-the-art VLMs.

2 Related Work

Datasets for Math OCR.
Most handwritten math OCR work has matured around the **CROHME** challenges, which evaluate symbol recognition and structure matching via Symbol Layout Trees (SLT) and ExpRate, but these benchmarks focus on *single expressions* only (Mouchère et al., 2016). Large-scale datasets such as **Im2LaTeX-100k** (Deng et al., 2017), **HME100K** (Yuan et al., 2022), and **MathWriting** (Gervais et al., 2025) extend the scale, yet they remain largely isolated-expression oriented. A first

step toward **multi-line evaluation** was introduced in the ICDAR 2023 **MLHMER** competition (Gao et al., 2023), but standardized document-level settings remain nascent.

Evaluation Metrics for Math OCR.

Text-based. String-level metrics like BLEU and edit distance are widely adopted in image-to-LaTeX tasks (Deng et al., 2017), but are brittle to format variations and fail to capture semantic equivalence.

Embedding-based. BERTScore (Zhang et al., 2020) and COMET (Rei et al., 2020) fail in mathematical contexts because they measure local textual similarity but do not verify step-by-step global reasoning. An output may look similar while containing an invalid transformation that breaks the derivation.

Rule/Structure-based. Symbol Layout Trees (Mouchère et al., 2016) represent symbols as labeled groups and spatial relationships as labeled directed edges. They can be used to determine equivalence in symbol-layout structure, but not on its own determine mathematical equivalence. In contrast, Computer Algebra Systems (Meurer et al., 2017) can check whether two terms are equivalent by transforming them into a common form. However, both fail to account for derivation fidelity, and cross-line consistency.

Image-based. Rendering-based comparisons (PSNR, SSIM, LPIPS) mitigate markup variance, and the recent Character Detection Matching (CDM) (Wang et al., 2025) evaluates recognition at the rendered character level. However, these approaches assume renderable outputs and remain limited to single-line formulas.

LLM Evaluators. Recent work demonstrates LLMs' effectiveness as evaluators for reasoning tasks, achieving strong human alignment across a range of tasks (Zheng et al., 2023; Fu et al., 2024; Chiang et al., 2024; Liu et al., 2023), including educational grading (Chiang et al., 2024; Choi et al., 2026).

However, no existing metric jointly addresses semantic correctness, multi-line coherence, and transcription faithfulness—the gap our work targets.

3 The PINK Score

While conventional metrics such as BLEU fail to capture semantic equivalence and the varying importance of differences, we first use an LLM-based auto-grader to assign semantic scores to both the oracle transcription of the solution path and the OCR transcription. These scores reflect the educational impact of OCR outputs more directly than lexical matching, since they evaluate whether the transcribed solution preserves the reasoning needed for downstream assessment. However, semantic auto-grading alone cannot account for over-correction, where an OCR output receives a higher score by implicitly fixing a student's mistake. To address this limitation, we propose the PINK (Penalized INK-based) Score, a two-stage framework illustrated in Fig. 2: the first stage obtains semantic scores via LLM-based auto-grading, and the second explicitly detects and penalizes over-corrections according to their severity.

3.1 Semantic Auto-Grading System

The first stage of PINK obtains a semantic evaluation of the VLM's OCR output. For this purpose, we employ GPT-5 ('gpt-5-2025-08-07') as our auto-grader system, leveraging the GPT family's capabilities in both evaluation tasks (Liu et al., 2023; Chiang et al., 2024) and mathematical reasoning (AIME, 2025; OpenAI, 2025).

This system assesses mathematical correctness and logical flow irrespective of surface text forms. It employs a process-oriented rubric grounded in educational assessment principles. Following the philosophy that evaluating only final answers overlooks crucial problem-solving aspects (Polya, 1945; Schoenfeld, 1985), we adopt partial-credit scoring similar to standardized tests like AP Calculus (College Board, 2023; Cambridge International Education, 2024). Our rubric decomposes solutions into five components assessing the entire problem-solving arc, each worth 20 points for a total of 100, as detailed in Fig. 2 (Step 2). The full pedagogical basis and rubric specification are provided in Section B.1; detailed prompts appear in Section G.

The auto-grader applies the same rubric to two solution texts, producing comparable scores:

- **Model Score** (S_{model}): Score for the VLM's OCR output
- **Oracle Score** (S_{oracle}): Score for the ground-truth transcription (including any student errors)

As illustrated in Fig. 2, when a student makes a sign error yielding 0 but the VLM "corrects" it to 2, our grader initially assigns a perfect score ($S_{\text{model}} = 100$), whereas the true student solution achieves a lower score $S_{\text{oracle}} = 70$. To validate that such values are robust to grader choice and inherent LLM variance, we later replicated the full pipeline with the open-source Qwen3-80B as an alternative judge, achieving near-identical VLM-

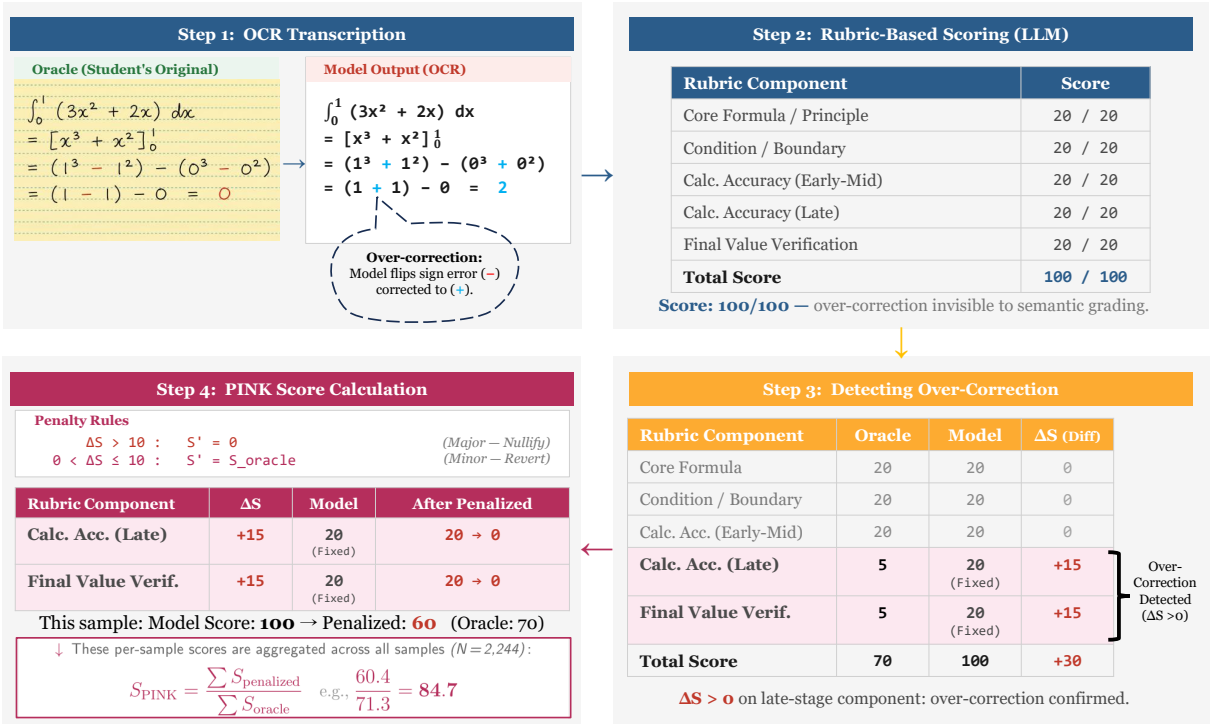


Figure 2: **Overview of the proposed PINK scoring system.** Step 1: a VLM OCR output over-corrects a student error. Step 2: the auto-grading system assigns a seemingly high score. Step 3: comparison with the oracle score reveals an over-correction event. Step 4: penalties are applied to compute the final PINK score.

level PINK scores (r=0.99, τ=0.90). Further analyses confirming stability across repeated runs and prompt variants appear in Section E.

3.2 Detecting Over-Correction

The second stage is designed to detect over-correction events that remain invisible to the auto-grader alone. Such situations are particularly problematic in educational contexts, where the goal is to preserve—not overwrite—the student's reasoning. In this context over-corrections are just as detrimental as wrongfully modified correct parts.

To capture this, we define over-correction events as cases where the model score exceeds the oracle score. For rubric component $r \in \{1, \ldots, R\}$:

$$\Delta S_r = S_{\text{model},r} - S_{\text{oracle},r} \tag{1}$$

An event occurs whenever $\Delta S_r > 0$. The example in Step 3 of Fig. 2 shows a $+15$ difference on the final answer rubric, flagging an over-correction. The magnitude of ΔS_r determines the penalty applied in the next stage.

3.3 Penalty Mechanism

Having identified over-correction, the next step is to decide how strongly it should be penalized. We adopt a two-tiered penalty system that separates minor variations from severe distortions, with the threshold set at 10 points—half the maximum score for any single rubric component.

Minor Over-Correction (Reversion): A score inflation of 1–10 points is considered a minor discrepancy. We attribute such differences to the probabilistic nature of LLM grading or trivial semantic variations that do not fundamentally alter the student's solution. The penalty is therefore a simple *reversion*: the component's score is replaced with the original oracle score $S_{\text{oracle},r}$.

Major Over-Correction (Nullification): A score inflation of 11 points or more exceeds the halfway mark for a rubric component (20 points). This indicates that the model has fundamentally altered or "corrected" a clear and significant error in the student's work. Such behavior is a serious distortion of the student's reasoning and is penalized with *nullification*, reducing the component's score to 0.

This logic is formally defined in the calculation of the penalized score for each rubric component:

$$S'_r = \begin{cases} S_{\text{model},r}, & \Delta S_r \leq 0, \\ S_{\text{oracle},r}, & 1 \leq \Delta S_r \leq 10 \quad \text{(Minor)}, \\ 0, & \Delta S_r > 10 \quad \text{(Major)}. \end{cases} \tag{2}$$

For a given test item, let R be the number of rubric components; the overall penalized score is:

$$S_{\text{penalized}} = \sum_{r=1}^{R} S'_r \tag{3}$$

In Step 4 of Fig. 2, two rubric scores are nullified from $20/20$ to $0/20$, lowering the overall score from 100 to 60. This ensures that inflated correctness is not misinterpreted as faithfulness.

Empirically, setting the threshold at $T{=}10$, which corresponds to half of the maximum rubric score (20 points), provides a robust balance between sensitivity and stability. Approximately 80% of all over-correction events fall below this boundary, while model ranking stability peaks near this value (Kendall's $\tau{>}0.9$ (Kendall, 1938)). These findings confirm that $T{=}10$ is both pedagogically meaningful and empirically optimal (see Section A for detailed ablation studies).

3.4 Final PINK Score Calculation

A raw penalized score alone can be misleading, since the oracle transcription in our perturbation dataset does not reach 100 (averaging 71.3). To obtain a comparable and interpretable metric, we normalize the penalized score by the oracle score.

Let N be the number of test items. For each test item n, let $S_{\text{penalized},n}$ and $S_{\text{oracle},n}$ denote its penalized and oracle scores, respectively. The final **PINK score** (S_{PINK}) is defined as

$$S_{\text{PINK}} = \frac{\sum_{n=1}^{N} S_{\text{penalized},n}}{\sum_{n=1}^{N} S_{\text{oracle},n}} \tag{4}$$

Under this definition, a model whose penalized scores equal the oracle scores achieves 1.0, and deviations from 1.0 quantify departures from faithful transcription.

4 Experimental Setup

Dataset We employ FERMAT (Nath et al., 2025), a dataset designed to assess VLMs' ability to detect, localize and correct mistakes in handwritten mathematical content. FERMAT is derived from 609 manually curated math problems. Based on a manually designed, comprehensive taxonomy of common student mistakes (computational, conceptual, notational, presentation, superficial), the authors adopted a human-in-the-loop approach to introduce targeted perturbations into correct solution paths. The resulting perturbed solution paths were transcribed by 43 human annotators, yielding 2,244 authentic handwritten samples. Notably, the original authors did not use FERMAT to explicitly assess OCR performance, and we are the first to adapt it for this purpose.

Baseline Models We selected 15 state-of-the-art VLMs with diverse architectures and scales (ranging from 7B to 124B) for a comprehensive performance comparison: Ovis2 (Lu et al., 2024), InternVL3 (Zhu et al., 2025), Qwen2.5-VL (Bai et al., 2025), LLaMA3.2 (Grattafiori et al., 2024), Pixtral (Agrawal et al., 2024), GPT-4o (OpenAI, 2023, 2024), and Gemini 2.5 Flash (Comanici et al., 2025). Full results for all models are reported in Table 1. We focus exclusively on VLMs, as established non-VLM OCR models are inapplicable to FERMAT: TrOCR lacks math-symbol coverage, Nougat cannot process handwriting, and Pix2tex is limited to single-line expressions. We also standardize the OCR prompt across all models, and prompt-sensitivity analyses show stable rankings under grading- and OCR-prompt paraphrases Sections E.3 and E.4.

Evaluation Metrics The primary metric used in our evaluation is the proposed **PINK score**. PINK measures semantic correctness while penalizing over-correction, providing a faithfulness-oriented alternative to conventional text-based metrics. For comparison, we also report **BLEU**, the most widely used baseline for math OCR, as well as **Edit Distance**, a standard string-level measure included for completeness. We omit Expression Rate (ExpRate), the standard metric for isolated formula recognition, as it is inapplicable to multi-line solutions mixing prose and equations and collapses to near 0% for all VLMs on FERMAT.

To support PINK computation, we also report the **Oracle Score** (S_{oracle}), the auto-grader's score on the ground-truth transcription. This baseline verifies that the perturbations reflect genuine mathematical errors and provides the normalization anchor used by PINK. The perturbed solutions average 71.3, while 500 clean samples score 95.1.

5 Results

5.1 Over-Correction is Pervasive and Systematic

Over-correction is pervasive: across all 15 evaluated VLMs, **42–66%** of transcriptions contain at

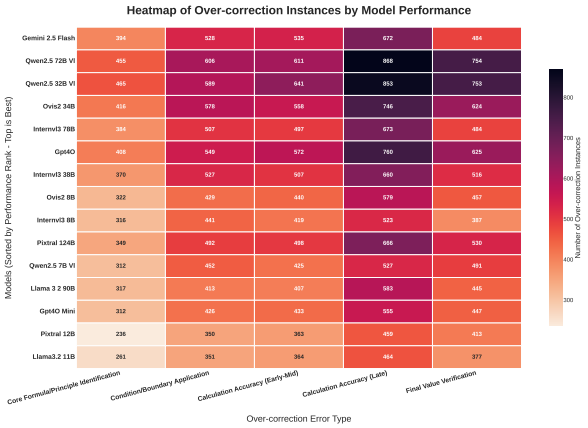


Figure 3: **Distribution of over-correction instances across rubric items for 15 models**, sorted by Penalized Score (Top is Best). Over-corrections are heavily concentrated in the final two stages, particularly in higher-performing models, consistent with autoregressive context accumulation overriding visual evidence.

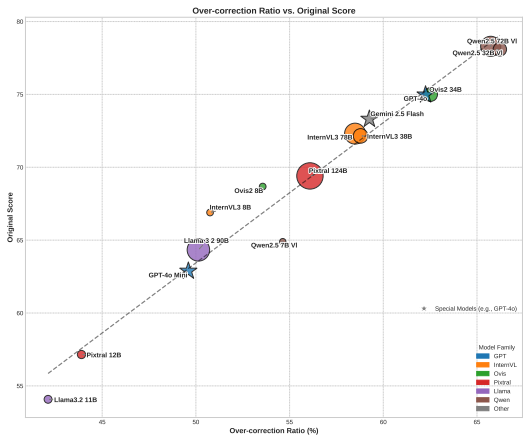


Figure 4: **Correlation between Original Score and Over-correction Ratio.** Bubble size represents the scale of the model.

least one instance where the model overwrites the student's original work. To understand this pervasive failure, we characterize the distribution and mechanism of this behavior below.

Where and When Does Over-Correction Occur? Our analysis reveals that over-correction is not a random perceptual glitch but a systematic behavior tied to the model's reasoning process. The heatmap in Fig. 3 exposes two prominent trends. First, over-corrections cluster heavily in the final stages of problem-solving—specifically in ***Calculation Accuracy (Late)*** and ***Final Value Verification***. Second, this concentration is markedly more pronounced in higher-performing models. This suggests that over-correction is an emergent property of sophisticated reasoning rather than a simple perceptual failure. As the model autoregressively processes a solution, it builds a strong, logically consistent internal narrative. When encountering visually inconsistent tokens (e.g., "5" in "$2 \times 3 = 5$"), these contradictions are often dismissed as noise. Larger and more capable models, which rely more heavily on their internal context, are therefore more likely to override visual evidence and produce logically inferred tokens instead.

Correlation with Model Scale and Properties. To investigate how model characteristics relate to over-correction, we analyze the bubble chart in Fig. 4. The chart plots each model's over-correction ratio against its rubric-based score, with bubble size indicating parameter count (see Section D.1 for full per-model frequencies).

A clear pattern emerges: **stronger models over-correct more**. Models achieving higher rubric-based scores also exhibit higher over-correction ratios. Over-correction is pervasive yet variable, ranging from **66.2%** (Qwen2.5 32B VL) to **42.1%** (Llama3.2 11B). High-performing models such as **Gemini 2.5 Flash** and **Qwen2.5 72B** cluster in the top-right region, reflecting both strong mathematical ability and a greater tendency to overwrite the student's work. The trend is consistent with a greater reliance on internal context, which can resemble the context-driven generation biases discussed in prior work (Ji et al., 2023).

5.2 Visual Evidence of Over-Correction

We analyze attention maps during over-corrected token generation (Fig. 5). When InternVL3-8B corrects a student's "sin" to "tan", attention on the handwritten error region drops sharply; a similarly sized faithful transcriber, Qwen2.5-VL 7B, maintains attention on the strokes. We observe similar patterns in additional cases, consistent with context accumulation attenuating visual grounding and motivating PINK's explicit penalty for score improvements over the oracle. Further details on attention extraction and case-level analysis are provided in Section C.3.

5.3 Standard Text Metrics Miss Over-Correction

Having established that over-correction is both pervasive and systematic, we now ask: do standard evaluation metrics reflect these differences in model faithfulness? We compare rankings under BLEU and PINK to test if a faithfulness-aware

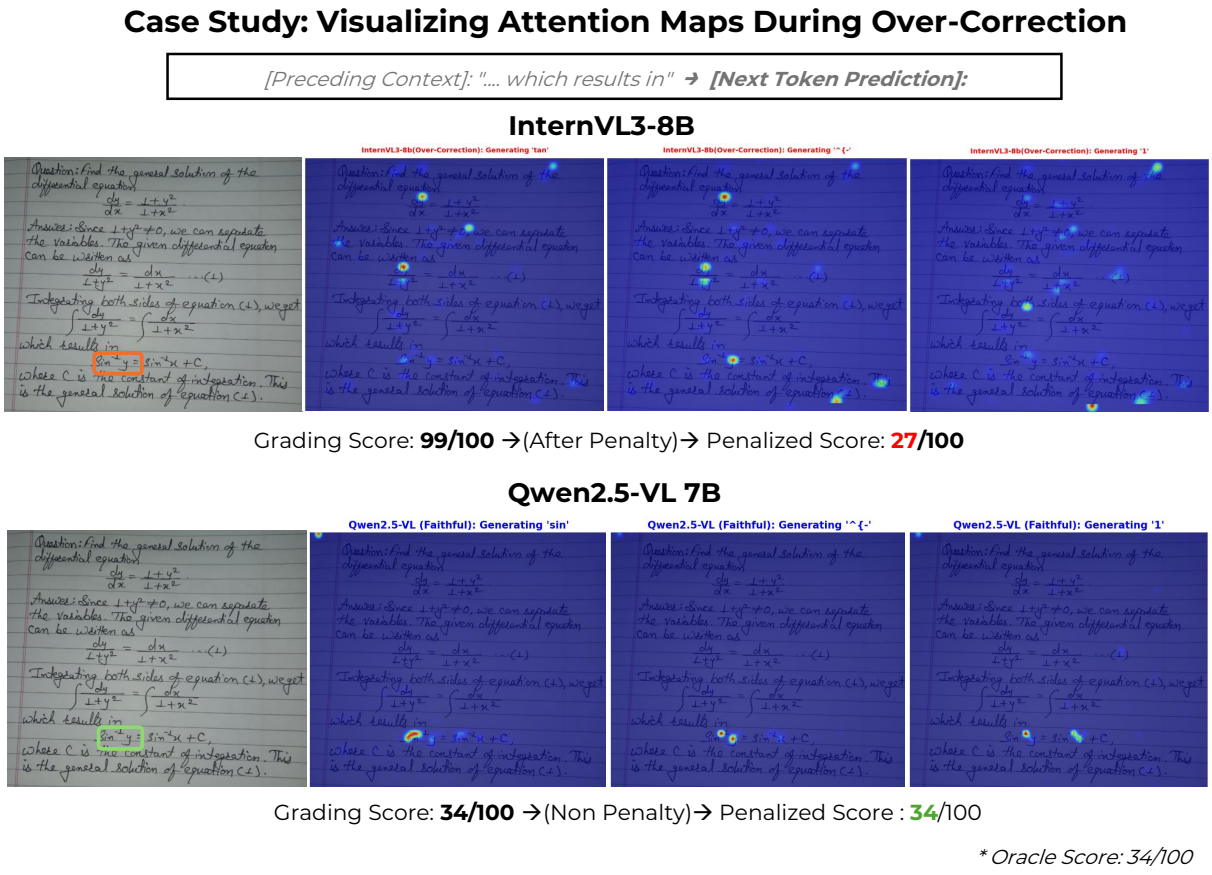


Figure 5: **Visualizing Attention Maps and Penalty Application.** We visualize the attention mechanism at the moment of generating the target token, conditioned on the preceding integral context. **(Top) InternVL3-8B** produces the mathematically consistent but unfaithful token "tan", while showing little attention to the handwritten "sin". Its high raw score of **99** is reduced to **27** after applying the PINK penalty. **(Bottom) Qwen2.5-VL-7B** faithfully preserves "sin" while maintaining attention on the relevant handwritten strokes. Its score remains **34**, matching the oracle score, with no penalty. This contrast is consistent with accumulated mathematical context attenuating visual grounding during over-correction.

metric yields meaningfully different assessments.

We compare the rankings of 15 open-source and proprietary VLMs on the FERMAT dataset (Nath et al., 2025) under two metrics: the conventional text-based BLEU score (Papineni et al., 2002) and our proposed PINK metric, which penalizes over-correction. The results in Fig. 6 are striking. An OCR leaderboard based on BLEU is dramatically overturned when re-evaluated with PINK. Models that achieve top ranks under BLEU, such as *GPT-4o* and *InternVL*, experience a substantial drop in rank. Notably, *Gemini 2.5 Flash*, previously a mid-tier model under BLEU, ascends nine places to claim the top position under PINK. This reversal provides strong evidence that over-correction is not a rare edge case, but a frequent failure mode in SoTA VLMs. Crucially, this divergence is not explained by rubric grading alone: under raw rubric scores the two models rank similarly (Gemini 5th, GPT-4o 4th). The gap emerges only when transcription faithfulness is accounted for (see Section D.2 for the full transition across all 15 VLMs).

To better understand this ranking reversal, we closely examine the two models with the largest discrepancy: *GPT-4o* and *Gemini 2.5 Flash*. The distribution plot in Fig. 7a shows the difference between the rubric-based score assigned to a model's OCR transcription and that assigned to the ground-truth transcription, where a positive value indicates over-correction. *GPT-4o* exhibits a **longer and heavier right tail** than *Gemini 2.5 Flash*, indicating both more frequent and more severe over-corrections. The scatter plot in Fig. 7b further supports this trend by comparing rubric-based scores before and after applying the over-correction penalty: *Gemini 2.5 Flash* remains tightly clus-

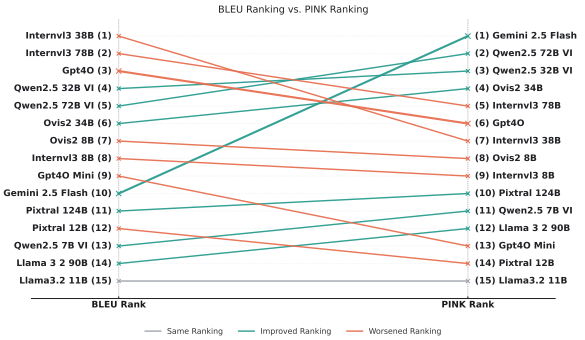


Figure 6: **Ranking Reversal of VLMs.** The leaderboard based on conventional BLEU scores (left) is dramatically overturned when re-evaluated with our PINK metric (right), which penalizes over-correction. Models like *GPT-4o* drop significantly, while *Gemini 2.5 Flash* ascends nine places to the top rank, showing that existing metrics miss the crucial dimension of faithfulness.

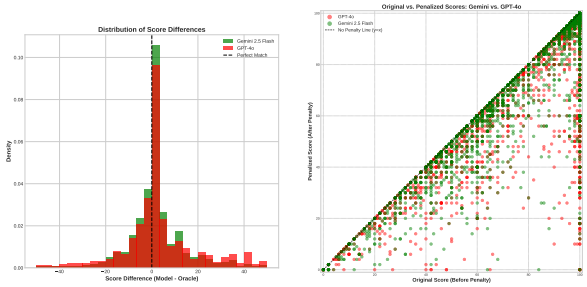
Table 1: **Performance comparison of open-source and closed-source models.** The table evaluates models using the PINK, BLEU, and Edit Distance scores. The background color intensity indicates the relative performance for each metric.

Model	PINK Score ↑	BLEU Score ↑	Edit Distance ↓
Open-source Models			
Qwen2.5 72B VL	0.920	0.733	0.1978
Qwen2.5 32B VL	0.919	0.734	0.1975
Ovis2 34B	0.916	0.723	0.1997
InternVL3 78B	0.911	0.749	0.1969
InternVL3 38B	0.905	**0.771**	0.1925
Ovis2 8B	0.887	0.652	0.2321
InternVL3 8B	0.873	0.642	0.2364
Pixtral 12B	0.850	0.606	0.2836
Qwen2.5 7B VL	0.832	0.565	0.3332
Llama-3.2 90B VI	0.814	0.512	0.3339
Pixtral 124B	0.722	0.606	0.2495
Llama-3.2 11B VI	0.676	0.407	0.4649
Closed-source Models			
Gemini 2.5 Flash	**0.935**	0.614	0.2007
GPT-4o	0.907	0.735	**0.1676**
GPT-4o Mini	0.787	0.618	0.2698

tered around the $y = x$ diagonal, while many outputs from *GPT-4o* fall far below it. Some *GPT-4o* samples drop from near 100 to near zero after penalization, showing that the model can produce logically consistent transcriptions that nevertheless severely distort the student's original work. For completeness, Table 1 reports the full quantitative results for all models across PINK, BLEU, and Edit Distance (Levenshtein, 1966), supporting the leaderboard reversal described above.

5.4 Human Evaluation

To validate whether PINK aligns better with human judgment than BLEU, we designed an evaluation



(a) Score Difference Distribution (b) Penalty Analysis Scatter Plot

Figure 7: **Over-Correction Rate**: Comparison between *GPT-4o* (red) and *Gemini 2.5 Flash* (green).

around two research questions:

- **RQ1 (Reliability):** How consistent is the LLM-based automated scoring system, which underpins the PINK score, with evaluations from human experts?
- **RQ2 (Preference):** Do human experts prefer the PINK score over BLEU for assessing the quality of OCR results?

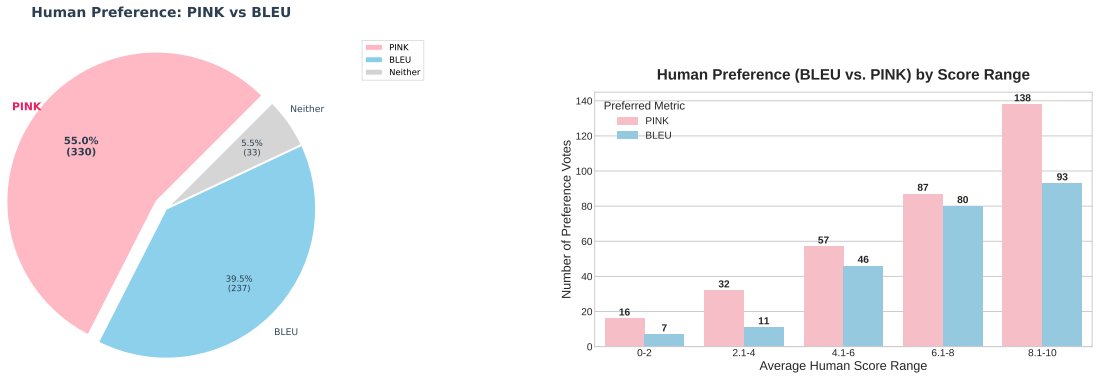
We recruited three expert annotators with university-level mathematical knowledge and an average of over two years of experience in grading mathematics assignments. The evaluation was performed on 200 samples extracted from the FERMAT dataset, transcribed by Ovis2-8B VLM. To ensure this sample was an unbiased representation, we confirmed its BLEU score distribution statistically mirrors the full dataset (mean 0.650 vs. 0.652). The annotators were assigned two tasks:

Direct Grading (for RQ1): For each of the 200 OCR outputs, annotators assigned an overall score on a 0-10 scale, guided by the same five evaluation categories used in our automated scoring system.

Preference Selection (for RQ2): This task directly measures which metric aligns better with human judgment. Annotators were first presented with a student's original handwritten solution and the correct solution. Then, the PINK and BLEU scores, calculated for a single OCR output of the student's work, were presented anonymously as 'Score A' and 'Score B'. The assignment of metrics to 'Score A' or 'Score B' was randomized for each sample to eliminate positional bias. Annotators then selected which score they believed better reflected the faithfulness of the OCR result.

The results of our evaluation are as follows.

RQ1: Auto-Grading System Correlates Strongly with Experts. To assess the reliability of our human evaluation, we measured inter-rater



(a) Human Preference: PINK vs BLEU.

(b) Preference analysis by score range.

Figure 8: **Human evaluation results.** (a) Experts showed a clear overall preference for PINK over BLEU. (b) PINK was consistently preferred across all score ranges, demonstrating its robustness.

reliability, with a Cohen's weighted kappa of 0.69, indicating a substantial level of agreement among annotators. We then compared the average expert scores with our LLM-based automated scores, obtaining a Pearson correlation coefficient of 0.78. These results support the use of automated scoring within PINK as a proxy for expert judgment.

RQ2: Experts Show a Clear and Consistent Preference for PINK over BLEU. As shown in Fig. 8a, experts more often judged PINK to better reflect OCR quality than BLEU. Overall, PINK was preferred in 55.0% of cases, compared to BLEU's 39.5% ($p < 0.001$, two-sided binomial test on item-level majority votes). The trend is not BLEU-specific: comparisons against MathBERTa (Novotný and Štefánik, 2022) and Edit Distance (Levenshtein, 1966) show consistent results (Section E.5). Experts also favored PINK across all score brackets (Fig. 8b), suggesting better alignment with perceived OCR quality. These findings support that penalizing over-correction improves alignment with expert judgment.

6 Conclusion and Future Work

Over-correction is not a rare edge case but a systematic failure mode affecting 42–66% of transcriptions across all 15 evaluated models, with stronger models over-correcting more frequently. Standard metrics like BLEU fail to detect this behavior; PINK, which explicitly penalizes over-correction, reverses model rankings and aligns better with human judgment (55% preference vs. BLEU's 39.5%). Our attention map analysis further suggests that autoregressive context accumulation attenuates visual grounding, indicating that stronger reasoning may come at the cost of transcription faithfulness—a trade-off that warrants further investigation. As VLMs are increasingly deployed in educational settings, ensuring that these models preserve rather than erase student reasoning must become a first-class evaluation criterion.

Our preliminary mitigation experiments reveal that over-correction is not easily resolved through prompting alone (Section F). While explicit instructions marginally reduced over-correction, they simultaneously degraded overall transcription accuracy, resulting in no net improvement. This suggests that over-correction is deeply entangled with the models' core reasoning capabilities, likely requiring architecture- or training-level interventions, such as targeted model editing approaches for LVLMs (Seong et al., 2025)

Limitations

Despite our extensive experiments and analyses, this work still has several limitations.

First, a high PINK score does not imply a universally superior VLM or OCR model. PINK specifically measures transcription faithfulness with respect to over-correction and should be interpreted as one evaluation axis rather than a holistic quality indicator. Second, our evaluation pipeline relies on an LLM-based auto-grader, with GPT-5 serving as the primary judge. This introduces a dependency on proprietary infrastructure that limits both reproducibility and accessibility. To mitigate this, we conducted cross-grader validation using the open-source Qwen3-80B, achieving strong agreement. We plan to release the complete evaluation codebase, including the open-source grading pipeline, to support independent replication. Third, all experiments are conducted on a single dataset, FERMAT.

This choice was driven by the current absence of alternative publicly available datasets that simultaneously feature multi-line, handwritten mathematical solutions with intentionally introduced student errors. Extending this evaluation across diverse data sources will be an important direction for future work. Finally, we acknowledge the limited scale of our human evaluation due to the difficulty of recruiting annotators with combined expertise in LaTeX, mathematics, and pedagogy. Nonetheless, the preference trends and score correlations strongly support PINK's alignment with expert judgment.

Our critique of over-correction should not be read as a negative assessment of VLM progress. The challenge lies in enabling models to distinguish when to reason and when to faithfully transcribe—we hope PINK can serve as a useful tool for navigating this trade-off.

Acknowledgements

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A Penalty Threshold Sensitivity

A.1 Distribution of Over-correction Magnitudes

We analyzed over-correction scores $\Delta S = S_{\text{model}} - S_{\text{oracle}}$ across all models on the FERMAT dataset. As shown in Table 2, 79.3% of all over-corrections fall within the 0–10 point range, with 60.4% concentrated below 5 points. Thresholds below 10 therefore over-penalize minor variations, while thresholds above 10 fail to capture significant over-corrections.

Table 2: Distribution of over-correction magnitudes across all models.

Range (points)	Occurrences	Percentage
0–5	22,483	60.4%
5–10	7,038	18.9%
10–15	3,548	9.5%
15–20	4,131	11.1%

A.2 Model-specific Behavioral Patterns

Figure 10 reveals how individual model rankings evolve across threshold values. Three distinct behavioral patterns emerge:

- **Robust models** (e.g., Gemini 2.5 Flash): Maintain top rankings regardless of threshold, demonstrating consistent faithful transcription
- **Minor over-correctors** (e.g., GPT-4o, Qwen2.5 VL family): Improve rankings at higher thresholds, indicating predominantly small-scale corrections ($\Delta S < 10$)
- **Critical over-correctors** (e.g., Ovis2 8B, InternVL3 family): Deteriorate under stricter thresholds, revealing frequent substantial alterations ($\Delta S > 10$)

Notably, Gemini 2.5 Flash's consistency may stem from its reasoning architecture, enabling better preservation of original content despite contextual pressure to correct errors.

B Detailed Methodology for Metric Design and Evaluation

In this section, we provide supplementary visual materials to further detail our evaluation framework, including the pedagogical basis for our rubric design and the user interface protocol used for human evaluation.

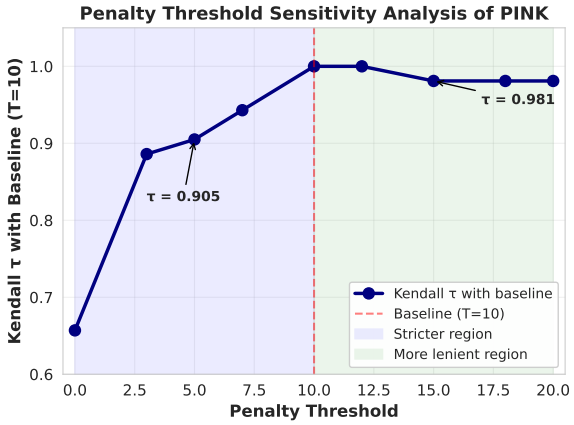

Figure 9: Ranking stability under different penalty thresholds T, where models are penalized when $\Delta S(S_{\text{model}} - S_{\text{oracle}}) > T$ (i.e., VLM score exceeds student's true score by more than T points). Kendall's τ measures ranking consistency relative to T=10 baseline. High stability ($\tau > 0.9$) in the range $T \in [5, 15]$ validates our threshold choice.

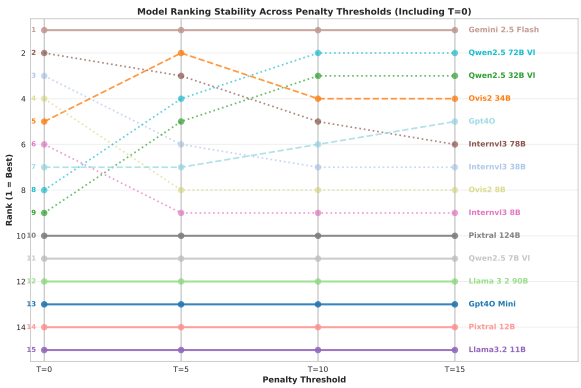

Figure 10: Model ranking trajectories under varying penalty thresholds. Three patterns emerge: robust (e.g., Gemini 2.5 Flash), minor over-correctors (e.g., GPT-4o), and critical over-correctors (e.g., InternVL3).

B.1 Pedagogical Basis and Rubric Specification

Our grading rubric is not arbitrarily defined but is rigorously grounded in process-oriented assessment principles from educational theory. Traditional binary scoring (correct/incorrect) has long been criticized for failing to reflect a learner's problem-solving ability, conceptual understanding, and strategic choices (Polya, 1945; Schoenfeld, 1985).

To address this, international standard assessments such as AP Calculus and Cambridge A-Level adopt a **Partial Credit** system. This system is structured around sequential problem-solving steps: "Problem Understanding → Condition Ap-

Rubric	Score	Description
Core Formula/ Principle Identification	2	Correctly identified and stated the necessary formulas for solving the problem **Example:** *For $\int_0^1 (3x^2 + 2x)\,dx$: Applying power rule for integration*
Condition/ Boundary Application	2	Accurately applied given conditions (boundaries, constraints, units, etc.) **Example:** *Evaluates bounds [0,1]: $F(1) - F(0)$ where $F(x) = x^3 + x^2$*
Calculation Accuracy (Early-Mid)	2	Intermediate calculations were performed without omissions or contradictions **Example:** $\int (3x^2 + 2x)\,dx = x^3 + x^2 + C$
Calculation Accuracy (Late)	2	Final transformations and simplifications were logical and error-free **Example:** $[x^3 + x^2]_0^1 = (1 + 1) - (0 + 0) = 2$
Final Value Verification	2	Final answer correctly incorporates all conditions and calculations **Example:** *Final answer is 2.*
Total Points	10	Sum of all rubric scores

Figure 11: Overview of Auto-Grading System's Rubrics

plication → Calculation Procedure → Result Verification" (College Board, 2023; Cambridge International Education, 2024). This structure is widely recognized for ensuring transparency in evaluation and aiding in the diagnosis of specific error types (National Council of Teachers of Mathematics, 2014).

Consistent with these standards, we designed a fixed 5-component rubric (Figure 11), allocating 20 points to each stage (Total 100). By adopting this established educational framework, our metric can systematically capture mathematical thinking processes and error patterns, mirroring the standards of human expert grading.

B.2 Human Evaluation Interface and Protocol

Figure 12 illustrates the interface used for our human expert study. The evaluation protocol was designed to simulate a realistic grading scenario while ensuring blind comparison:

1. **Simulated Grading Scenario:** Annotators are presented with a math problem and a "Student Solution." Although this text is the VLM's OCR transcription, it is presented as a student's answer. Annotators are instructed to act as math teachers and grade the solution based on the provided Reference Answer.

2. **Blind Metric Comparison:** After assigning their own score (0-10 scale), annotators are shown two automated scores labeled **"Score A"** and **"Score B."** These correspond to the PINK and BLEU scores (normalized to the same scale). Crucially, the assignment of A/B is **randomized** for each sample to prevent positional bias and ensure that annotators remain unaware of which metric is which.

3. **Preference Selection:** Finally, experts select which score (A or B) better reflects the true quality and faithfulness of the transcription compared to their own expert judgment.

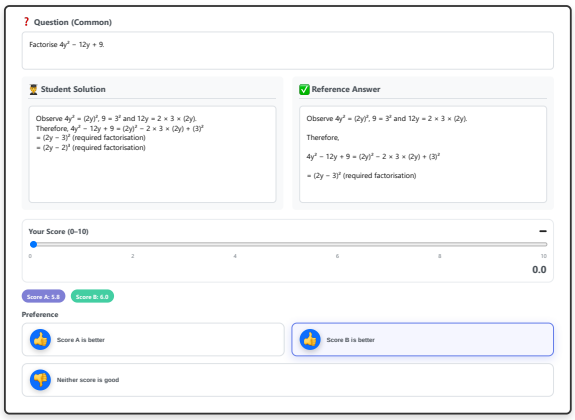

Figure 12: Human Evaluation: Labeling Tool

Also, annotators were recruited from our institution and compensated through regular employment. They were fully informed about the research objectives and how their evaluations would be used.

C Qualitative and Mechanistic Analysis of Over-Correction

To provide a qualitative understanding of how Vision-Language Models (VLMs) fail to faithfully transcribe student errors, we present a detailed analysis from two perspectives. First, we classify Over-Correction into two distinct categories based on severity—**Simple-Level** and **Critical-Level**—and illustrate how these behaviors artificially inflate scores using real examples from the FERMAT dataset (Fig. 13). Second, we go beyond phenomenological categorization to empirically validate the underlying mechanism. By visualizing the model's attention maps, we demonstrate that Over-Correction is driven by the model's internal reasoning overriding visual perception (Fig. 5).

C.1 Type 1: Simple-Level (Local Fix)

This type occurs when the model acts like a **mathematical spell-checker**, performing localized token substitutions to fix specific errors while preserving the overall structure.

- **Scenario:** The student incorrectly wrote $\sin^{-1}$ instead of $\tan^{-1}$ (Top of Fig. 13).
- **Model Action:** InternVL3-8B detects the inconsistency with the integral context ($\int \frac{dx}{1+x^2}$) and locally fixes the student's error by swapping "sin" for "tan".
- **Consequence:** The student's misconception is masked. A failing grade (Oracle: **34**) is inflated to a near-perfect score (**99**).

C.2 Type 2: Critical-Level (Global Rewrite)

A more severe failure mode where the model **discards the student's reasoning** and generates a correct summary based on its own internal calculation.

- **Scenario:** The student failed to derive the correct solution for an exponential system (Bottom of Fig. 13).
- **Model Action:** GPT-4o ignores the erroneous derivation steps entirely (visualized as the "Missing Area" in the figure) and simply outputs the correct final value: "On solving... we get x=4".
- **Consequence:** The evidence of the student's error is erased. A low score (Oracle: **14**) jumps to a perfect score (**100**).

C.3 Detailed Analysis of Visual Grounding

To empirically validate our hypothesis that **Over-Correction** is an emergent behavior where a model's internal reasoning suppresses visual evidence, we visualized the attention maps of two distinct models: **InternVL3-8B** (prone to over-correction) and **Qwen2.5-VL 7B** (a faithful transcriber).

We hypothesize that as a VLM autoregressively generates the solution, the accumulation of mathematical logic creates **"contextual pressure."** When the visual evidence (e.g., a student's error) conflicts with this internal context, the model treats the visual tokens as **noise** and suppresses them, prioritizing its internal prediction instead.

To prove this, we analyzed a case where a student incorrectly wrote $\sin^{-1} y = \sin^{-1} x + C$ following an integral $\int \frac{dx}{1+x^2}$. We extracted the attention weights at the exact moment the model generates the target token, given the full history of tokens the model had generated autoregressively up to that point.

As illustrated in Figure 5, the results confirm our hypothesis and justify the necessity of the PINK penalty mechanism:

- **InternVL3-8B (Over-Correction):** Under the pressure of the **autoregressively accumulated context**, the model generates the mathematically correct token "tan". Crucially, the attention map reveals **negligible activation** on the visual region corresponding to the handwritten "sin", indicating that the model treated the visual evidence as noise. While this "correction" yields a high initial grading score of **99/100**, it falsely inflates the student's ability. Consequently, PINK identifies this unfaithful behavior and applies a penalty, drastically reducing the final score to **27/100**.
- **Qwen2.5-VL 7B (Faithful Transcription):** In contrast, this model faithfully transcribes the student's error ("sin"). The attention map demonstrates strong and precise activation on the handwritten strokes, showing that it successfully resisted contextual pressure to maintain visual grounding. Although the resulting score is lower (**34/100**), it accurately reflects the student's actual performance (matching the Oracle baseline), thus incurring **no penalty**.

These visualizations provide direct evidence that Over-Correction allows reasoning to override perception. This confirms that high raw scores in reasoning models can be misleading, validating PINK's penalty mechanism as essential for ensuring evaluation integrity.

D Over-Correction: Frequency and Ranking Impact

D.1 Per-Model Over-Correction Rates

Fig.14 illustrates that over-correction is a pervasive issue, yet its prevalence varies significantly, ranging from **66.2%** in Qwen2.5 32B VI down to **42.1%** in Llama3.2 11B.

D.2 Ranking Reshuffle via Penalty

The application of our over-correction penalty dramatically reshapes the raw auto-graded leaderboard, as shown in Fig. 15. This shift exposes a core limitation of relying solely on the initial

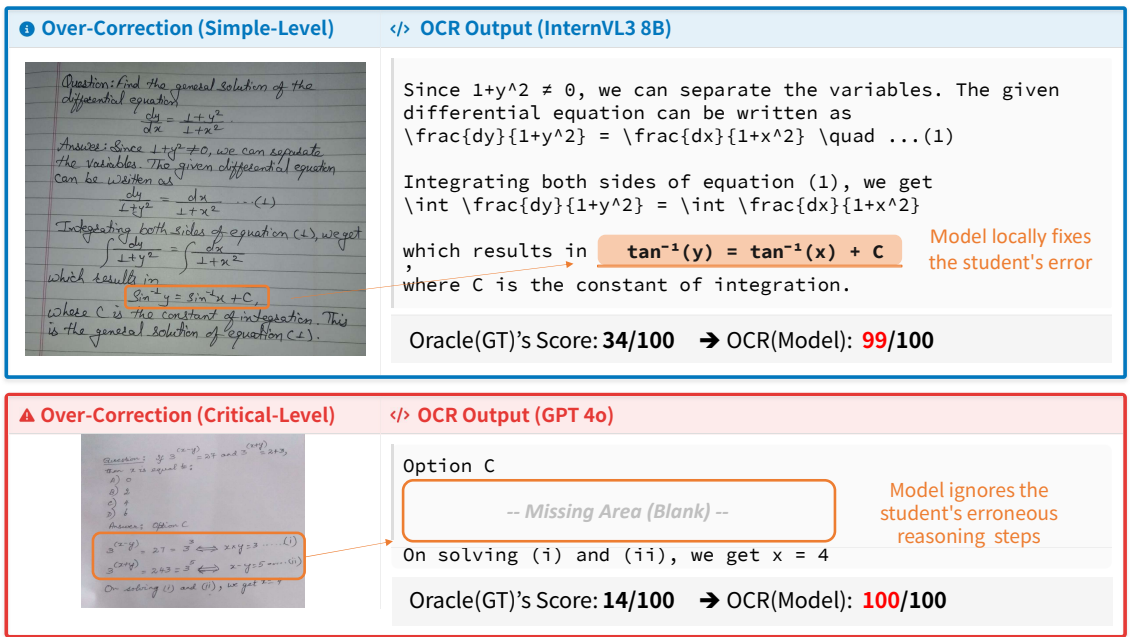

Figure 13: **Case Study of Over-Correction. (Top) Simple-Level:** InternVL3-8B locally fixes the student's error ($\sin \rightarrow \tan$), inflating the score from 34 to 99. **(Bottom) Critical-Level:** GPT-4o ignores the student's erroneous reasoning steps and provides a correct summary, inflating the score from 14 to 100.

auto-graded scores: they fail to distinguish between faithful transcription and cases where the model quietly "fixes" student errors.

A clear example is the contrast between **Gemini 2.5 Flash** and **GPT-4o**. Gemini rises from **5th** under the raw auto-graded ranking to **1st** after the penalty is applied, while GPT-4o drops from **4th** to **6th** due to frequent over-corrections. This reversal highlights that PINK surfaces fidelity failures that the raw scores overlook, yielding a model ranking that better reflects true OCR behavior.

E Robustness and Reproducibility

We validate reproducibility by comparing **Qwen3-Next-80B-A3B-Thinking** (open-source) against GPT-5 across 15 VLMs and 33K+ samples.

E.1 Cross-Grader Consistency Analysis

As summarized in Table 3, our analysis confirms that PINK is grader-agnostic.

Despite moderate sample-level variance in grading (r=0.78) and over-correction penalization (r=0.74), **aggregation into PINK scores yields near-perfect agreement (r=0.99)**, resulting in highly consistent VLM rankings (τ=0.90). Notably, both graders identify the *same* top-tier VLMs (Gemini, Qwen2.5, Ovis2 34B) and bottom-tier VLMs (Llama-11B, Pixtral-12B). We will release the Qwen3-80B grading toolkit to enable full community reproduction.

Stage	Level	Pearson r	QWK	Kendall's τ
Grading	Sample	0.78	0.78	–
OC Penalty	Sample	0.74	0.73	–
PINK Score	VLM	**0.99**	–	–
Final Rank	VLM	–	–	**0.90**

Table 3: **Reproducibility Pipeline.** Micro-variances at the sample level cancel out, yielding near-perfect macro agreement. N = 33k (sample), N = 15 (VLM).

E.2 Run-to-Run Grading Stability

To quantify the effect of LLM grading variance on PINK, we graded GPT-4o's OCR outputs five times independently using the open-source Qwen3-80B grader, which exhibits near-perfect agreement with GPT-5 (r = 0.99, Table 3). We analyze N = 1,803 samples with successful grades across all runs. The resulting PINK scores yield a coefficient of varia-

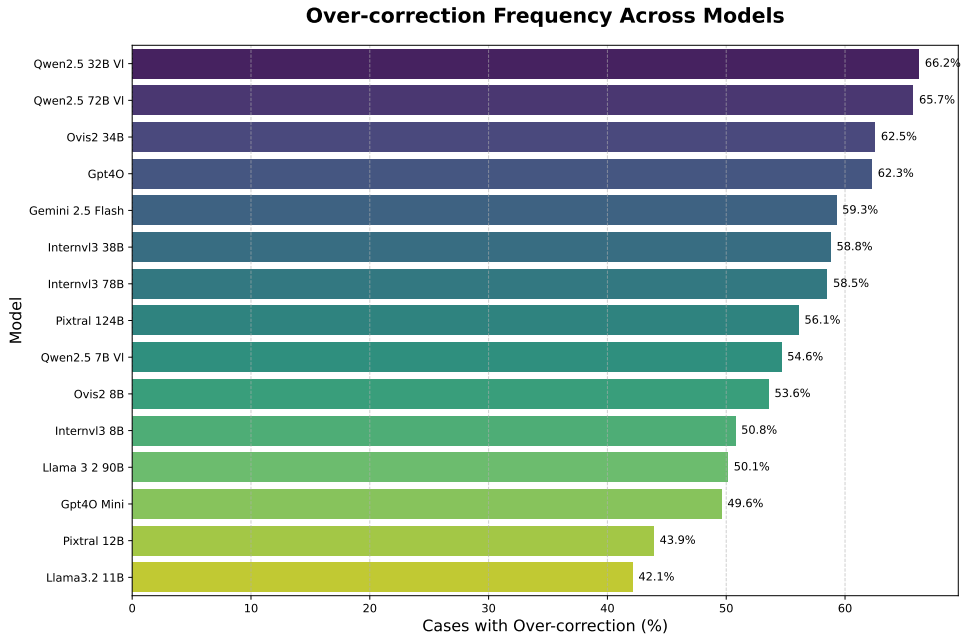

Figure 14: Over-correction frequency across all 15 tested VLM models.

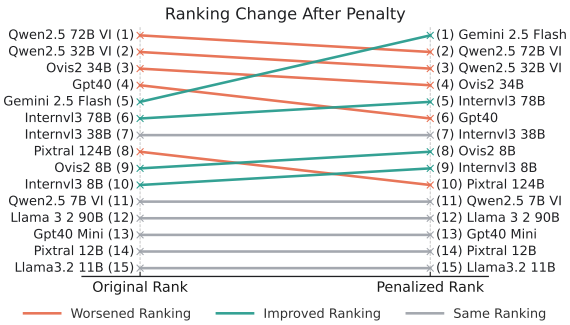

Figure 15: **Ranking changes after applying the over-correction penalty.** The PINK score leads to a significant reordering of models, highlighting the impact of penalizing unfaithful corrections.

Model	PINK Score		Rank	
	GPT	Qwen	GPT	Qwen
Gemini 2.5 Flash	**0.94**	0.87	1	3 ↓
Qwen2.5-72B-VL	0.92	0.88	2	2 -
Qwen2.5-32B-VL	0.92	**0.88**	3	1 ↑
Ovis2 34B	0.92	0.85	4	4 -
InternVL3-78B	0.91	0.84	5	6 ↓
GPT-4o	0.91	0.85	6	5 ↑
InternVL3-38B	0.91	0.83	7	7 -
Ovis2 8B	0.89	0.80	8	8 -
InternVL3-8B	0.87	0.79	9	9 -
Pixtral Large	0.85	0.79	10	10 -
Qwen2.5-7B-VL	0.83	0.73	11	12 ↓
Llama 3.2 90B	0.81	0.73	12	11 ↑
GPT-4o Mini	0.79	0.69	13	13 -
Pixtral 12B	0.72	0.60	14	14 -
Llama 3.2 11B	0.68	0.58	15	15 -

Table 4: **Leaderboard Stability.** Comparison using GPT-5 vs. Qwen3-Next-80B-A3B-Thinking judges.

Table 5: **Run-to-run stability.** GPT-4o graded 5× by Qwen3-80B ($N = 1{,}803$).

Metric	Value
PINK Mean (5 runs)	0.872
PINK CV	0.41%
Penalized Score QWK	0.926

tion (CV) of 0.41%, with pairwise agreement on penalized scores of QWK = 0.926 (Table 5). This confirms that sample-level grading noise is effectively absorbed by aggregation.

E.3 Auto-Grader's Prompt Sensitivity

To assess whether PINK scores depend on specific rubric wording, we created three prompt variants by modifying key phrases (e.g., "character by character" → "line by line"), yielding an average BLEU-1 of 0.86 relative to the original prompt. We evaluated four representative VLMs—one per PINK quartile—using Qwen3-80B as the grader. As shown in Table 6, all models maintain identical rankings across variants, with PINK deviations of std ≤ 0.006, confirming that our metric is robust to prompt wording.

E.4 OCR Prompt Sensitivity

To assess whether cross-model comparisons depend on the specific OCR prompt wording, we created three paraphrased variants of the original OCR prompt and evaluated four representative VLMs—one per PINK quartile—on a subset of 1,500 samples, with all outputs graded by Qwen3-80B-. We

Table 6: PINK scores across three rubric prompt variants. Grader: Qwen3-80B. Models selected from each PINK quartile.

VLM (Quartile)	Orig	V1	V2	V3	Std
Gemini 2.5 Flash (Q1)	.873	.875	.870	.877	±.003
GPT-4o (Q2)	.853	.856	.857	.864	±.004
InternVL3-8B (Q3)	.787	.792	.793	.802	±.005
Llama-3.2-11B (Q4)	.582	.592	.593	.596	±.005

selected different representative models from Section E.3 to increase coverage across architectures. As shown in Table 7, the quartile ordering is preserved across all prompt variants (Q1 > Q2 > Q3 > Q4). The inter-quartile gaps ($\geq .07$) consistently exceed within-model prompt variation (std $\leq .015$). Weaker models exhibit slightly higher sensitivity, but not enough to alter the quartile ordering, indicating that cross-model PINK comparisons are robust to OCR prompt paraphrases.

Table 7: PINK scores across OCR prompt variants (graded by Qwen3-80B, $N = 1{,}500$). Quartile ordering is preserved across all variants.

VLM (Quartile)	Orig	V1	V2	V3	Std
Qwen2.5-VL-32B (Q1)	.873	.872	.872	.877	±.002
Ovis2-8B (Q2)	.804	.807	.799	.810	±.005
Qwen2.5-VL-7B (Q3)	.725	.738	.729	.731	±.005
LLaMA-3.2-11B (Q4)	.617	.654	.628	.634	±.015

E.5 Extended Baseline Comparisons

To verify that PINK's advantage over BLEU is not specific to a single baseline, we conducted two additional human preference studies following the same protocol as Section 5.4 (200 items × 3 annotators, majority vote per item). We compared PINK against MathBERTa, a BERT model pretrained on mathematical text including LaTeX formulas, and Edit Distance. MathBERTa-F1 was rescaled with a domain-specific baseline (F1 = 0.8786, estimated from 1,000 random OCR–reference pairs) and mapped to the same 0–10 scale used for PINK. As shown in Table 8, PINK was consistently preferred over all three baselines, with all differences statistically significant. Table 9 further shows that the three similarity-based metrics form a highly correlated cluster ($\tau \geq 0.75$), while all diverge substantially from PINK ($\tau \leq 0.52$), confirming that the ranking gap reflects a class-level limitation of similarity-based metrics rather than a BLEU-specific artifact.

Table 8: Human preference: PINK vs. three baselines (200 items, 3 annotators). p: two-sided exact binomial test on non-tied majority-vote items.

Baseline	PINK	Baseline	Tied	p
MathBERTa	124 (62%)	52 (26%)	24 (12%)	5.75×10^{-8}
Edit Dist.	122 (61%)	57 (29%)	21 (11%)	1.33×10^{-6}
BLEU	119 (60%)	69 (35%)	12 (6%)	3.27×10^{-4}

Table 9: Kendall's τ rank correlation across four metrics (15 VLMs).

	MathBERTa	Edit Dist.	BLEU	PINK
MathBERTa	1.00	0.75	0.88	0.47
Edit Dist.	0.75	1.00	0.88	0.52
BLEU	0.88	0.88	1.00	0.50
PINK	0.47	0.52	0.50	1.00

F Prompt-Based Mitigation Attempt

To investigate whether over-correction can be reduced at inference time, we augment the OCR prompt with explicit faithfulness instructions (e.g., "Transcribe EXACTLY what is written, including any errors"; see Figure 19 for the full prompt). We select six representative open-source VLMs spanning the PINK range: three high-performing models (Qwen2.5-32B-VL, Qwen2.5-72B-VL, Ovis2-34B) and three lower-performing models (Qwen2.5-7B-VL, Pixtral-12B, Llama-3.2-11B) from Table 4. Closed-source models (Gemini 2.5 Flash, GPT-4o) were excluded as their system prompts cannot be freely modified. All outputs are graded by the open-source Qwen3-80B judge, which exhibits near-perfect agreement with the proprietary GPT-5 judge ($r = 0.99$, $\tau = 0.90$; Table 3), on $N = 2{,}207$ samples where grading succeeded for all six VLMs under both prompts (37 of 2,244 excluded due to Qwen3-80B grading failures). Absolute values may therefore differ slightly from Table 4; all within-table comparisons remain internally consistent.

The mitigated prompt reduces over-correction frequency by 4 percentage points on average (Table 10), confirming that VLMs do respond to faithfulness instructions. However, this improvement comes at a clear cost for stronger models: the three high-PINK models show *lower* PINK scores after mitigation ($\overline{\Delta} = -0.02$), because the conservative transcription strategy also suppresses correct mathematical content.

The one exception is Llama-3.2-11B, where OC % slightly increases (+3) yet PINK improves

Table 10: **Prompt-based mitigation.** OC: samples with ≥1 over-corrected item. Δ: mitigated − original. Green: improved; Red: degraded.

	PINK Score			OC (%)		
Model	Org	Mit	Δ	Org	Mit	Δ
High-PINK (open-source top-3)						
Qwen2.5-32B	.88	.85	−.03	62%	53%	−10
Qwen2.5-72B	.87	.84	−.03	61%	50%	−11
Ovis2-34B	.85	.85	.00	59%	54%	−4
Low-PINK (open-source bottom-3)						
Qwen2.5-7B	.73	.74	+.01	46%	46%	0
Pixtral-12B	.59	.61	+.02	41%	38%	−3
Llama-3.2-11B	.57	.64	+.07	39%	42%	+3
Avg	.75	.75	+.01	51%	47%	−4

substantially (+.07). We attribute this to the faithfulness prompt improving overall transcription completeness for this weaker model, even though a few additional over-correction events are introduced.

Overall, the net effect on PINK is negligible ($\overline{\Delta} = +.01$), demonstrating that **prompt-level intervention alone is insufficient** to resolve over-correction without sacrificing transcription quality. This motivates architectural or training-level approaches as future work.

G Prompts

To ensure full reproducibility and transparency of our evaluation framework, we provide the exact prompts utilized in our experiments. These prompts were carefully engineered to enforce strict output formats (e.g., JSON) and to align the LLM's behavior with our specific evaluation criteria.

- **Auto-Grading Prompt (Figure 16):** This prompt instructs the LLM judge (GPT-5) to act as a rigorous math exam grader. It embeds the 5-component rubric detailed in Section B and enforces a structured JSON output containing both scores and textual justifications for every deduction.

- **Auto-Labeling Prompt (Figure 17):** Used for error analysis, this prompt guides the LLM to distinguish between semantic recognition errors (e.g., wrong numbers) and cosmetic formatting differences (e.g., LaTeX spacing). This separation is crucial for quantifying how often BLEU penalizes correct answers due to formatting.

- **OCR Prompt (Figure 18):** The standardized prompt used across all 15 baseline VLMs. It explicitly instructs models to transcribe handwritten content into LaTeX format without adding conversational filler, ensuring a fair comparison.

- **Mitigated OCR Prompt (Figure 19):** A variant of the OCR prompt augmented with explicit faithfulness instructions (e.g., "transcribe exactly as written, including errors"). This prompt is used in the mitigation experiment described in Section F.

Auto-Grading Prompt

Role:

```
You are a math exam grader. Evaluate the student solution using the provided rubric and grading guidelines.
```

Rubric Structure (Scale 100):

```
{
  "rubric": [
    {
      "item": "Core Formula/Principle Identification",
      "points": 20,
      "criterion": "Correctly identified and stated the necessary formulas for solving the problem"
    },
    {
      "item": "Condition/Boundary Application",
      "points": 20,
      "criterion": "Accurately applied given conditions (boundaries, constraints, units, etc.)"
    },
    {
      "item": "Calculation Accuracy (Early-Mid)",
      "points": 20,
      "criterion": "Intermediate calculations were performed without omissions or contradictions"
    },
    {
      "item": "Calculation Accuracy (Late)",
      "points": 20,
      "criterion": "Final transformations and simplifications were logical and error-free"
    },
    {
      "item": "Final Value Verification",
      "points": 20,
      "criterion": "Final answer correctly incorporates all conditions and calculations"
    }
  ],
  "notes": "Points may be deducted for unit, rounding, or notation errors"
}
```

Points may be deducted for unit, rounding, or notation errors

Grading Guidelines:

1. **Analyze each expression character by character** to identify any errors in formulas, calculations, or transformations, and provide a detailed justification explaining exactly where and why points were awarded or deducted.
2. Include specific numbers, equations, or character positions in justification when necessary (e.g., "In line 3, the expression 2x+3 should be 2x-3, where the + sign at position 3 is incorrect").
3. Total score must be an integer between 0-100.
4. Output only the JSON format below. No additional text.

Output Format:

```
{
  "scores": [
    {"item": "Core Formula/Principle Identification", "score": 0~20, "justification": "..."},
    {"item": "Condition/Boundary Application", "score": 0~20, "justification": "..."},
    {"item": "Calculation Accuracy (Early-Mid)", "score": 0~20, "justification": "..."},
    {"item": "Calculation Accuracy (Late)", "score": 0~20, "justification": "..."},
    {"item": "Final Value Verification", "score": 0~20, "justification": "..."}
  ],
  "total":
}
```

Figure 16: Auto-Grading Prompt

Auto-Labeling Prompt

```
You will be given two texts:
1. GT (Ground Truth): The original correct LaTeX mathematical expression
2. Prediction: The OCR-predicted LaTeX mathematical expression

Your task is to:
1. Identify all format differences using the categories below
2. Identify all recognition errors using the types below
3. Note the specific locations and nature of differences for the analysis
4. Classify based on what you found:
   - If only format differences found → "LaTeX format difference only"
   - If only recognition errors found → "Recognition error only"
   - If both found → "Both LaTeX and recognition errors"
   - If neither found → "No errors"


Format Difference Categories:


- Math Mode Difference: $ x $ vs \( x \) vs x
- Spacing Difference: \quad vs \qquad vs regular space
- List Creation Method: \begin{enumerate} vs 1. ... 2. ...
- Fraction Style: \frac{a}{b} vs a/b
- Relation Symbols: \le vs \leq vs <=
- Arrow Difference: \to vs \rightarrow vs ->
- Vector Notation: \vec{v} vs \mathbf{v}


Recognition Error Types:


- Wrong symbol: OCR read + instead of -, x instead of y
- Wrong number: OCR read 3 instead of 8
- Wrong word: OCR read "solve" instead of "solution"
- Missing content: OCR missed symbols, numbers, or words
- Extra content: OCR added content not in GT
- Broken LaTeX: Malformed LaTeX commands


Tolerance Rules (NOT errors):


- Extra or missing spaces (except in LaTeX commands)
- Case differences in regular text
- Consistent variable case changes
- Minor punctuation spacing


Output Format:
```

```
{
  "category": "LaTeX format difference only" | "Recognition error only" | "Both LaTeX and recognition errors" | "No errors",
  "format_differences": ["list of format differences found"],
  "recognition_errors": ["list of recognition errors found"],
  "gt_has_math_error": 0 | 1,
  "analysis": "Detailed explanation of ALL differences found"
}
```

Figure 17: Auto-Labeling Prompt

OCR Prompt

System Prompt:

```
You are a math assistant specializing in extracting mathematical question and answer content from handwritten images of math problems by middle or high school students. Your task is to analyze the given Image, which contains the handwritten math Question-Answer pair, and convert it into LaTeX format. Follow the specific instructions provided carefully and output your response strictly in the requested format. Ensure that your evaluation is precise, accurate, and based solely on the information visible in the image.
```

User Prompt:

```
The image provided contains a handwritten math problem with both a Question and an Answer at a middle or high school level. Your task is to explicitly perform OCR on the handwritten text and extract the content in LaTeX format. Return only the extracted content, formatted in LaTeX, exactly as it appears in the image, in the **Question:** and **Answer:** fields.

Ensure that no extra information is added that is not in the image.

Please return the LaTeX output as follows:
**Question:**
**Answer:**
```

Figure 18: OCR Prompt

OCR Prompt (Mitigated Ver.)

System Prompt:

```
You are a math assistant specializing in extracting mathematical question and answer content from handwritten images of math problems by middle or high school students. Your task is to analyze the given Image, which contains the handwritten math Question-Answer pair, and convert it into LaTeX format. Follow the specific instructions provided carefully and output your response strictly in the requested format. Ensure that your evaluation is precise, accurate, and based solely on the information visible in the image.

IMPORTANT: You must transcribe EXACTLY what is written, including any errors or mistakes made by the student. Do NOT correct or fix any mathematical errors.
```

User Prompt:

```
The image provided contains a handwritten math problem with both a Question and an Answer at a middle or high school level. Your task is to explicitly perform OCR on the handwritten text and extract the content in LaTeX format.

CRITICAL INSTRUCTIONS FOR FAITHFUL TRANSCRIPTION:
- Transcribe EXACTLY what is written in the image, including any mathematical errors or mistakes.
- Do NOT correct, fix, or improve any errors in the student's work.
- If the student wrote "2 + 2 = 5", you must transcribe it as "2 + 2 = 5", NOT "2 + 2 = 4".
- Your goal is FAITHFUL TRANSCRIPTION, not mathematical correction.
- Preserve all calculation errors, wrong answers, and mistakes exactly as they appear.

Return only the extracted content, formatted in LaTeX, exactly as it appears in the image, in the **Question:** and **Answer:** fields.

Ensure that no extra information is added that is not in the image.

Please return the LaTeX output as follows:
**Question:**
**Answer:**
```

Figure 19: Mitigated OCR Prompt